Cyber Secure – The Human Factor in Cybersecurity Project

Addressing & Mitigating the Weakest Link In Organisational Security

By George Nicholl

Overview

Cyber Secure is a browser-based Cybersecurity Awareness training platform designed to address human-driven security risks within organisations. The system focusses on educating non-technical users on common cybersecurity threats through structured, accessible learning modules that promote safer digital behaviour in both personal and workplace environments.

The platform targets widely recognised human-factor vulnerabilities such as phishing attacks, weak password practices, unintentional data sharing, neglecting software updates, and security fatigue. Rather than relying on policy-heavy or compliance-based training, Cyber Secure delivers concise, practical guidance that users can easily understand and apply.

This project was developed as part of a final-year cybersecurity project and demonstrates the ability to design, build and evaluate a complete security-focussed system. It combines front-end web development and user-centred design principles and real-world evaluation to validate effectiveness.

Cyber Secure showcases skills in system design, secure thinking, accessibility-focussed development, structured documentation and evaluation of system-impact – all within the context of a real and relevant cybersecurity problem.

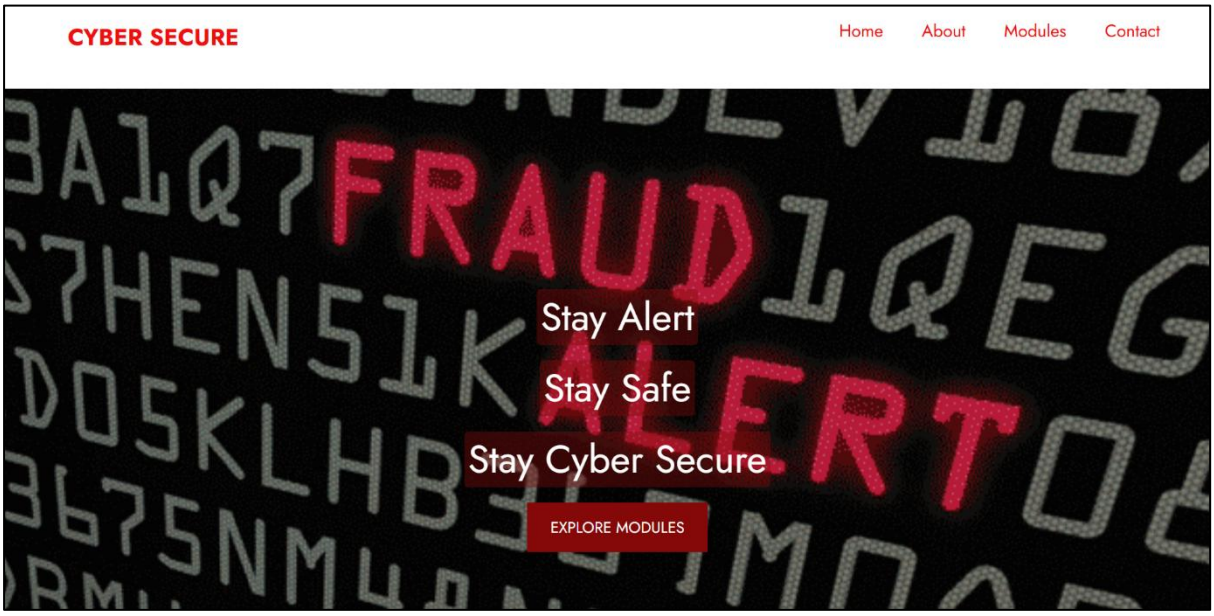

Figure 1: Cyber Secure Homepage (Final Version)

Technology Stack

The Cyber Secure Platform was developed using a lightweight and accessible front-end technology stack. This approach was chosen to ensure broad compatibility, ease of deployment, and suitability for environments with restrictive IT policies, such as corporate or educational networks.

The system does not rely on external frameworks or backend services in its current implementation, allowing it to run entirely within a standard web browser without additional configuration.

Component	Description
Languages Used	HTML5, CSS3, JavaScript
Frameworks / Libraries	None (Vanilla front end development)
Development Tools / IDE	Visual Studio Code
Design Approach	Modular, user-centred, accessibility-focussed
Architecture Type	Static, multi-page web application
Browser Compatibility	Fully Compatible with modern browsers (Chrome, Edge, Firefox)
Deployment Model	Local hosting or static web hosting

The decision to use a static front-end architecture was intentional. By avoiding backend dependencies and databases, the platform remains portable, low maintenance, and secure by design. This approach also reduces potential attack surfaces while ensuring that the tool can be easily deployed in environments where installing additional software or services may not be permitted.

HTML5 was used to structure all pages using semantic markup, allowing learning modules and supporting pages to share a consistent layout. CSS3 controls the visual presentation of the platform, with an emphasis on readability, spacing and visual consistency. Responsive design principles were applied to ensure usability across different screen sizes.

JavaScript was used sparingly to support basic interaction and demonstration functionality. This restrained use of scripting avoids unnecessary complexity and reduces potential security risks while still allowing for future expansion if required.

Hosting and Deployment Considerations

Cyber Secure was designed to be deployable using static hosting methods. Because the platform does not rely on server-side processing or databases, it can be hosted on any standard web server or static hosting provider with minimal configuration.

This allows the system to be deployed locally, within internal organisational networks, or on public static hosting platforms. The absence of backend dependencies reduces maintenance overhead and limits potential attack surfaces, making the platform suitable for training environments with strict security requirements.

Project Goals and Learning Outcomes

Project Goals

The primary goal of the Cyber Secure Project was to design and implement an accessible cybersecurity awareness platform that addresses common human-factor vulnerabilities within organisations. The system was intended to improve awareness of everyday cyber risks while remaining usable by individuals with little or no technical background.

A key objective was to move away from traditional compliance-based training and instead deliver content that is practical, relatable and easy to retain. The platform was also designed to be deployable in a wide range of environments without requiring complex infrastructure or configuration.

Learning Outcomes

Through the development of Cyber Secure, a range of technical and professional skills were demonstrated. These include designing user-centred security education tools, translating cybersecurity concepts for non-technical audiences, and structuring modular web-based systems.

The project also strengthened skills in documentation, evaluation, and critical reflection. Real user feedback and testing were used to assess system effectiveness and guide refinement, demonstrating an ability to evaluate solutions beyond basic functionality.

System Design and Architecture

Cyber Secure follows a static, multi-page web architecture designed to prioritise security, portability, and ease of deployment. Each page within the system represents either a learning module or a supporting component, with shared navigation and styling ensuring consistency across the platform.

A static architecture was deliberately chosen to minimise complexity and reduce security risk. By avoiding server-side processing, databases and persistent user sessions, the platform removes common attack vectors such as injection attacks, misconfigured authentication mechanisms, and data leakage. This design decision allows Cyber Secure to deliver its core educational purpose without introducing unnecessary technical risk.

All pages are implemented using static HTML files, with a shared CSS stylesheet providing consistent layout and visual presentation. This approach ensures predictable behaviour across environments and simplifies maintenance, as updates to styling or structure can be applied globally,

From a user perspective, the architecture supports both linear and non-linear learning paths. Users may progress through modules sequentially or select topics based on individual relevance, allowing flexibility while maintaining a clear and intuitive navigation structure.

The system architecture also supports future extensibility. While the current implementation does not include backend services, the modular separation of pages and components allows

server-side functionality – such as authentication, progress tracking, or analytics to be integrated in future versions without requiring a complete redesign.

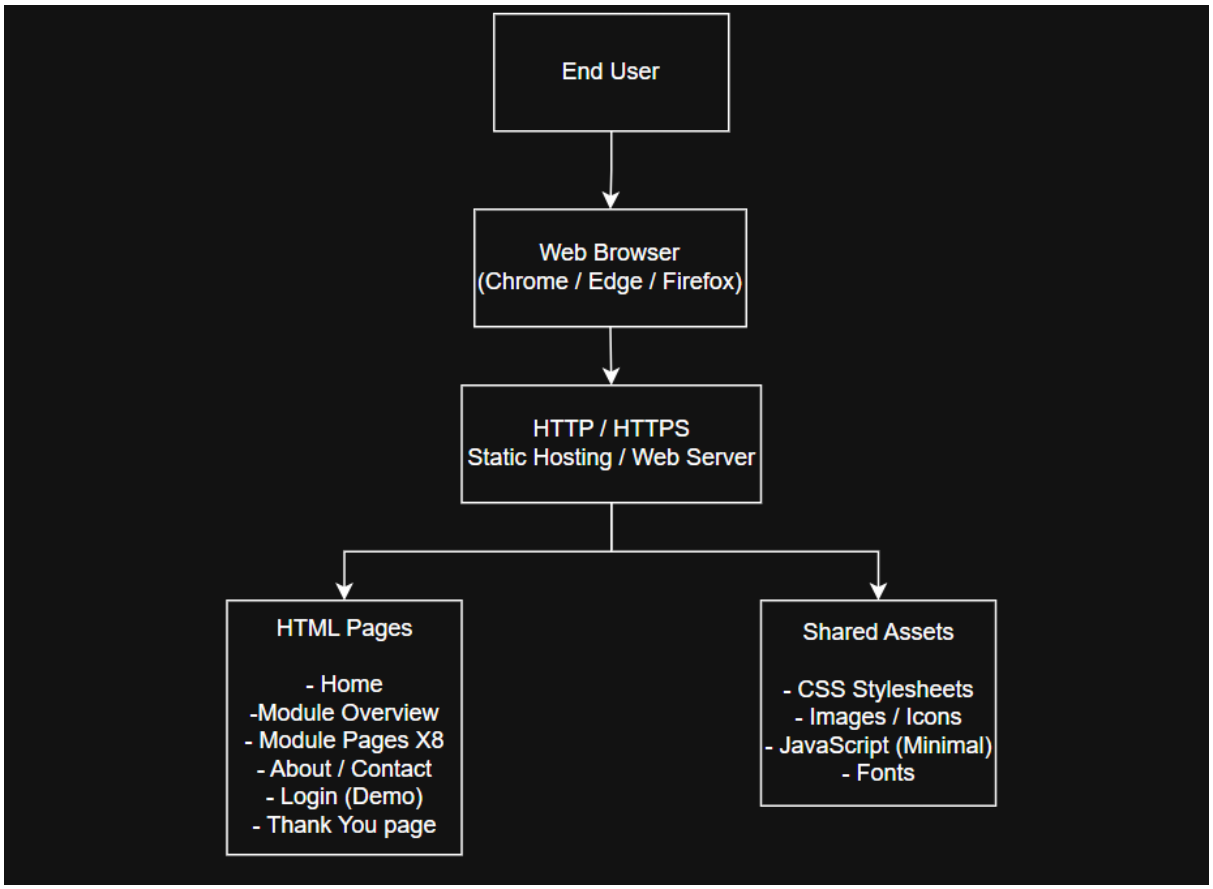

Figure 2: High Level Architecture of Cyber Secure Platform

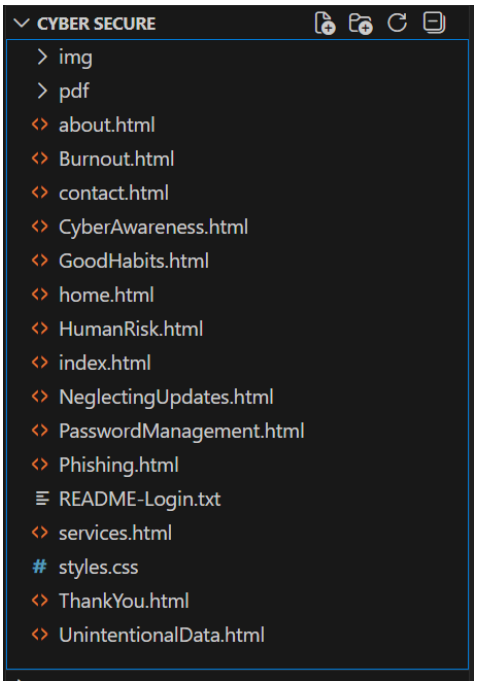

Figure 3: Project Folder Structure

Core Components

Homepage & Navigation

The homepage acts as the central entry point to Cyber Secure. It introduces the purpose of the platform and establishes context for users who may have limited prior exposure to cybersecurity concepts. The layout is intentionally simple, using clear headings and calls to action to guide users toward the learning modules.

Navigation elements are consistent across all pages, reducing cognitive load and allowing users to focus on content rather than interface complexity. This consistency is particularly important for non-technical users, who may disengage if navigation is unclear or unpredictable.

Learning Modules

The learning modules form the core functionality. Each module targets a specific human-factor cybersecurity risk, including phishing, password management, unintentional data sharing, neglecting software updates, general cyber awareness, human risk, burnout and good security habits.

Modules were selected based on their prevalence in real-world security incidents and their relevance to non-technical users. Rather than presenting abstract technical concepts, each module focuses on realistic scenarios that are likely to encounter in everyday digital activity.

All modules follow a consistent structure consisting of an introduction to the threat, real-world context, and practical mitigation guidance. This structured approach reinforces learning by establishing familiarity across modules and allowing users to focus on content rather than adapting to new formats.

Login Page (Demonstration)

A static Login page is included as a demonstration component to illustrate how authentication could be incorporated into a future version of the platform. The page does not process or store credentials and does not provide access control in the current version.

Including this component demonstrates foresight in system design while avoiding the risks associated with implementing incomplete or insecure authentication logic. It highlights how the platform could evolve into a personalised training system without compromising the security or integrity of the current version.

Supporting Pages

Supporting Pages such as About, Contact, and Thank You contribute to the complete and professionalism of the platform. The About page provides context and intent, the Contact page offers a point of communication, and the Thank You page reinforce completion and progress after engagement with learning content.

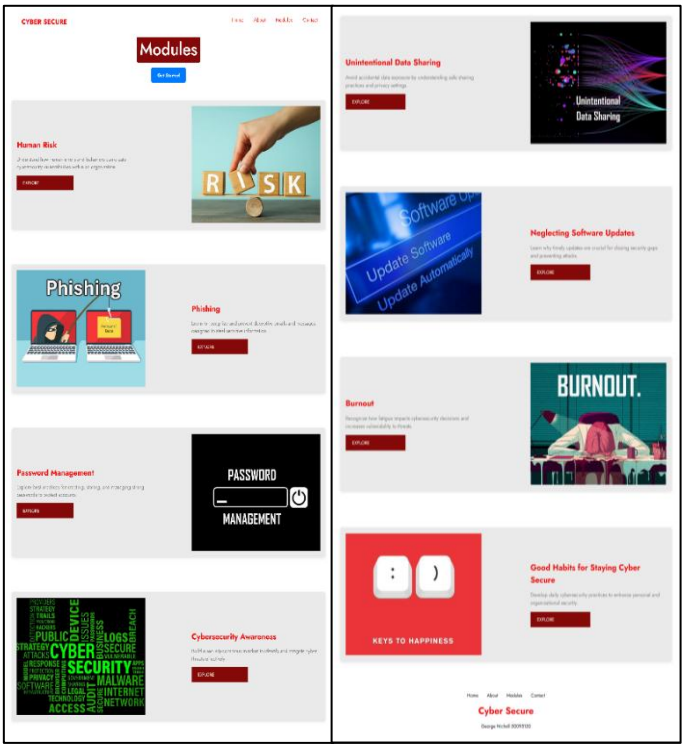

Figure 5: Modules Overview Page

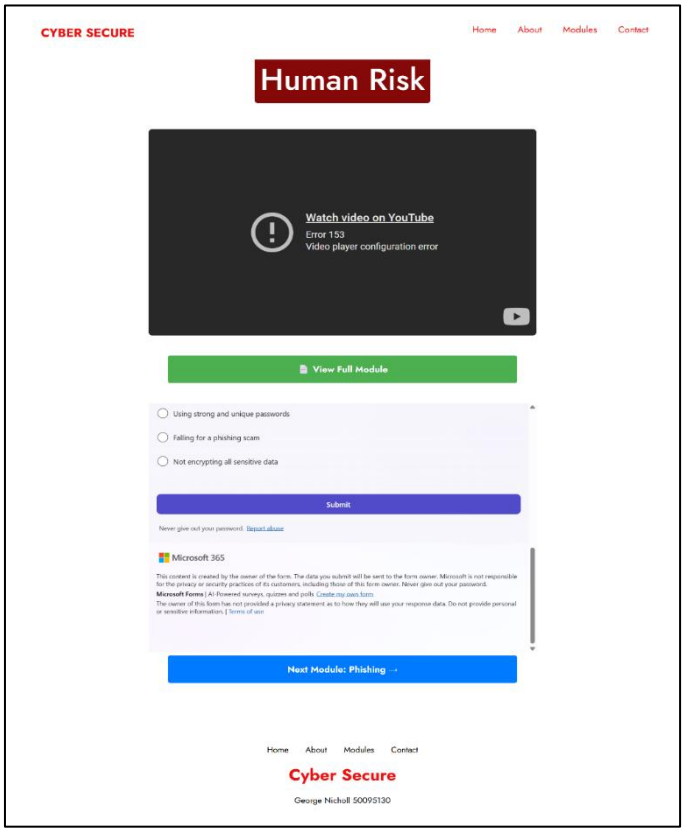

Figure 4: Example Module Page (Human Risk)

Figure 6: Login Page Demo

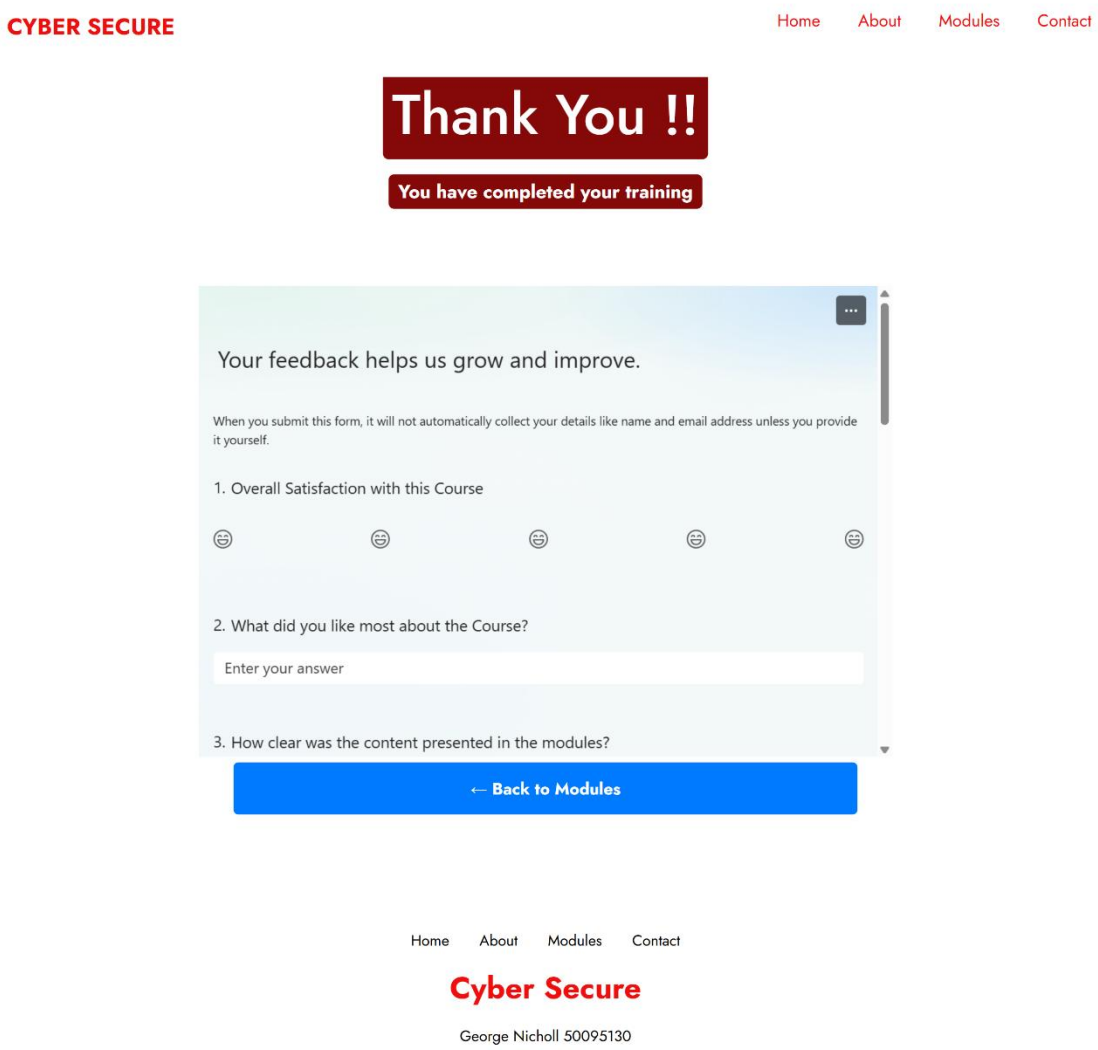

Figure 7: Thank You page

Front-End Implementation

HTML5 was used to structure all pages using clear and semantic markup. Each page follows a consistent layout, allowing shared styling and predictable navigation across the platform.

CSS3 defines the visual appearance of the system. Styling decisions prioritised readability, spacing and accessibility. A consistent colour scheme and typography were applied across all pages, with responsive design ensuring usability on different devices.

JavaScript usage was intentionally minimal. It supports basic interaction and demonstration features without introducing unnecessary complexity or security risk.

Device Compatibility and Responsiveness

Cyber Secure was designed to be usable across a range of devices, recognising that cybersecurity training is often completed in flexible environments rather than at a fixed workstation. Users may access the platform on desktop computers, laptops or mobile devices, depending on organisational policies and personal preference.

Page layouts were structured to adapt naturally different screen sizes without requiring separate versions of the site. Content areas resize and reflow to maintain readability, ensuring that learning material remains clear and accessible on smaller screens. Navigation elements were kept consistent and unobtrusive to avoid disrupting the learning experience on touch-based devices.

Text sizing, spacing and contrast were considered to support accessibility and reduce visual fatigue, particularly when modules are completed on mobile devices. Key content was tested at different screen widths to confirm that headings, paragraphs and interactive elements remained usable without horizontal scrolling or layout breakage.

By prioritising device compatibility during development, Cyber Secure supports remote learning scenarios and bring-your-own-device requirements while maintaining a consistent user experience across platforms.

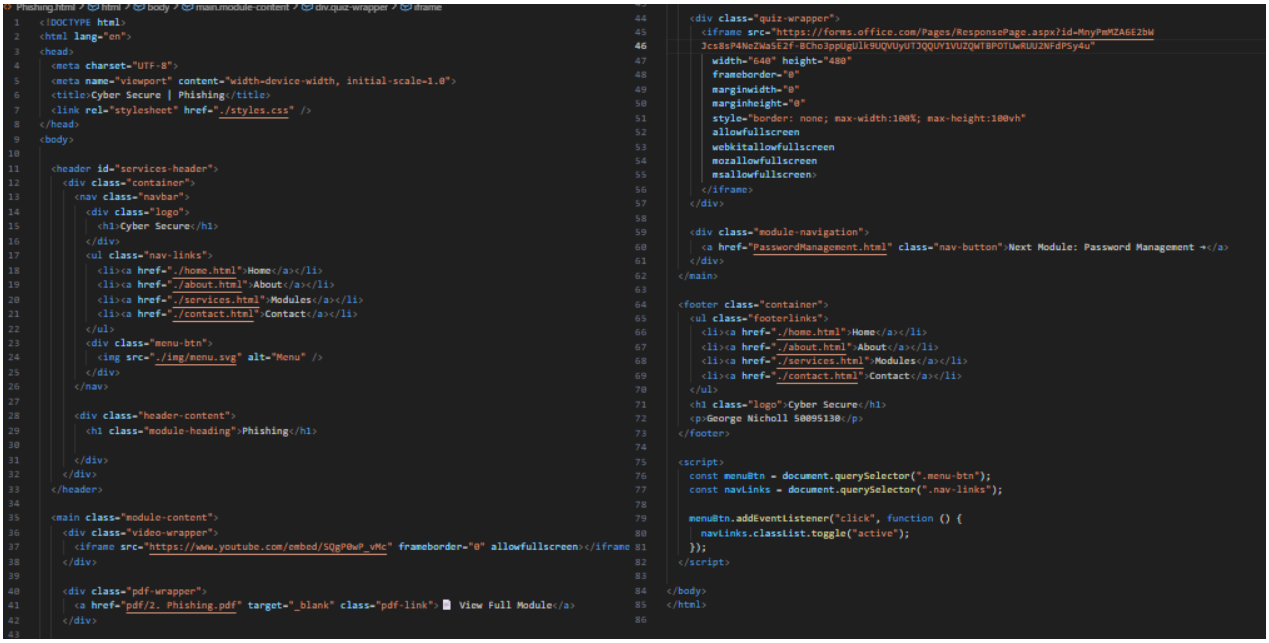

Figure 8: HTML Structure of a Module (Phishing)

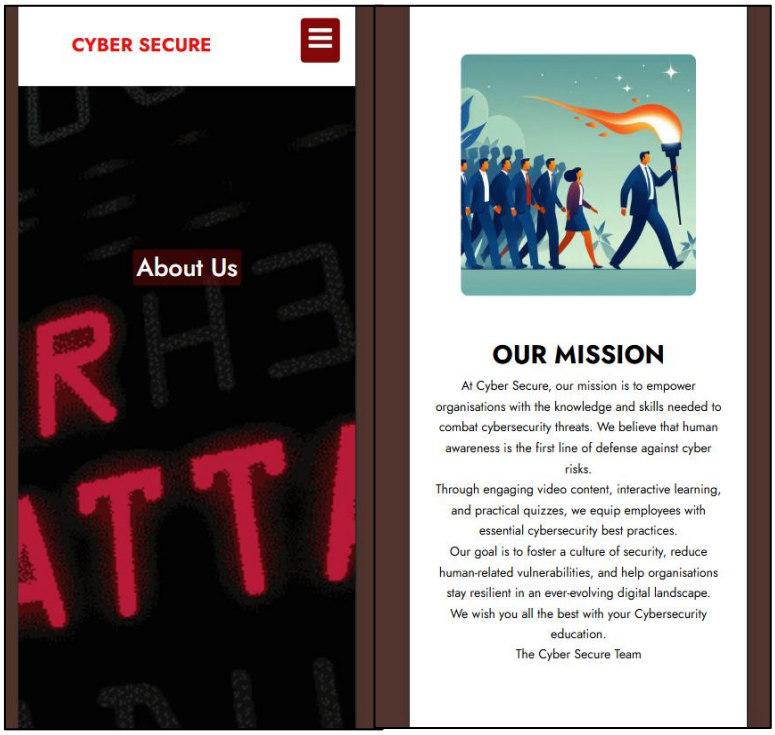

Figure 9: Responsive Layout on a Mobile Device

Development Process

The development of Cyber Secure followed an iterative process. An initial prototype was created to validate structure and content, followed by user feedback and self-evaluation. Based on this feedback, refinements were made to improve visual consistency, navigation flow and mobile responsiveness.

Content with key module were expanded to include clearer explanations and more relatable examples. A login demonstration page was added in the final development phase to better represent how the platform could function in a real deployment.

Time constraints required prioritisation of core functionality over advanced features, resulting in a focused and polished final version.

Figure 10: Early Prototype Version

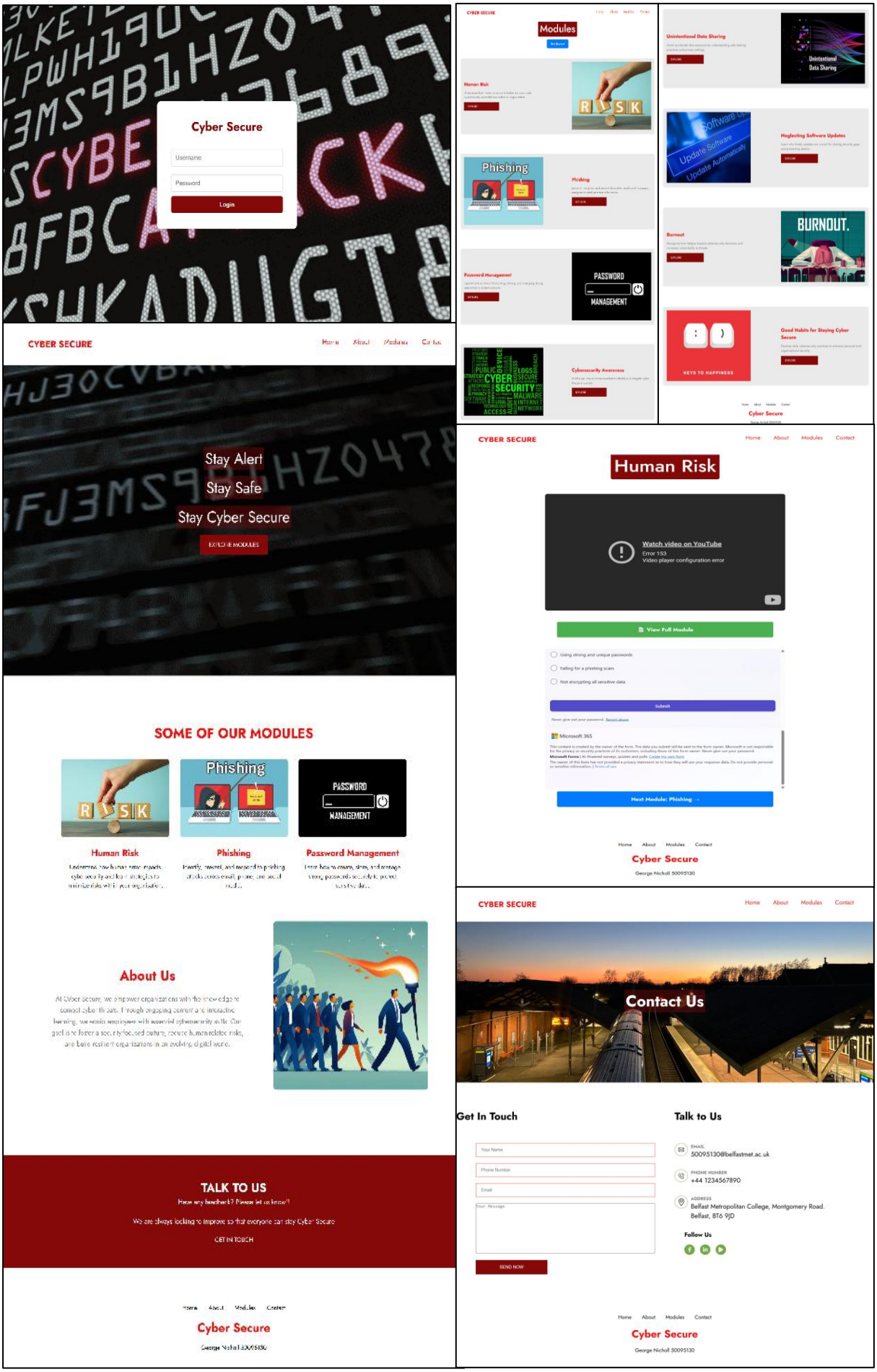

Figure 11: Final Version of Platform

Testing & Evaluation

Cyber Secure was evaluated through a combination of functional testing, user-based evaluation, and knowledge assessment. Functional testing focussed o ensuring that all pages loaded correctly, navigation links behaved as expected, and layout remained consistent across devices and screen sizes.

User Evaluation was conducted with participants from non-technical backgrounds to assess clarity, usability and perceived usefulness. Feedback indicated that users found the platform easy to navigate and reported increased confidence in recognising common cyber threats after completing the modules.

Knowledge-based assessments were used to measure understanding before and after engagement with the platform. Results showed a clear improvement in user awareness following module completion, with high levels of short-term knowledge retention observed during follow-up assessment.

While long-term knowledge retention was limited by project scope and timescales, the evaluation demonstrated that Cyber Secure was effective in improving awareness and understanding among its intended audience. These findings validate the platforms design approach and support its suitability as an organisational awareness tool.

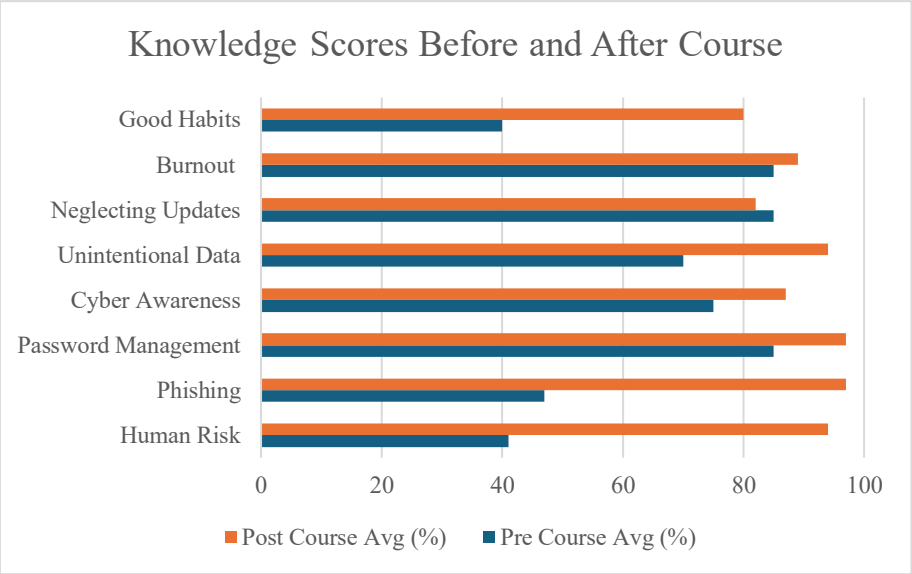

Figure 12: Summary of Evaluation Results

Challenges & Solutions

Several challenges were encountered throughout the development. Time constraints required careful prioritisation of core functionality over advanced features, influencing architectural and design decisions.

Participant availability during evaluation limited the volume of long-term retention data that could be collected. This was mitigated by focusing on clarity of content and short-term validation, while identifying extended evaluation as an area for future work.

A key challenge involved balancing functionality with security. Implementing dynamic features such as user accounts and progress tracking would have increased system complexity and risk. To address this, a static architecture was adopted, prioritising reduced attack surface and ease of deployment while maintaining clear pathways for future expansion.

Future Improvements

Future iterations of Cyber Secure could include a backend integration to support user accounts, progress tracking and personalised feedback. Additional enhancements such as gamification, automated reminders, and enterprise deployment options could further increase engagement and its effectiveness.

The platform also has potential to be adapted into a mobile application or integrated into existing organisational training systems.

Conclusion

Cyber secure demonstrates the design and implementation of a complete cybersecurity awareness training platform focussed on addressing human-factor risks. Through clear structure, accessible content, and real-world evaluation, the system shows how security education can be delivered effectively to non-technical users.

This project highlights skills in system design, front-end development, secure thinking, evaluation and professional documentation, making it a strong portfolio piece that reflects real-world considerations and deployment awareness.